

Master Advanced Prompting Techniques



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Overview



Chain-of-thought Prompting

Knowledge Augmentation

- Relevant keywords
- Structured data
- Additional context

Prompt Tuning

- Prompt tuning vs prompt engineering

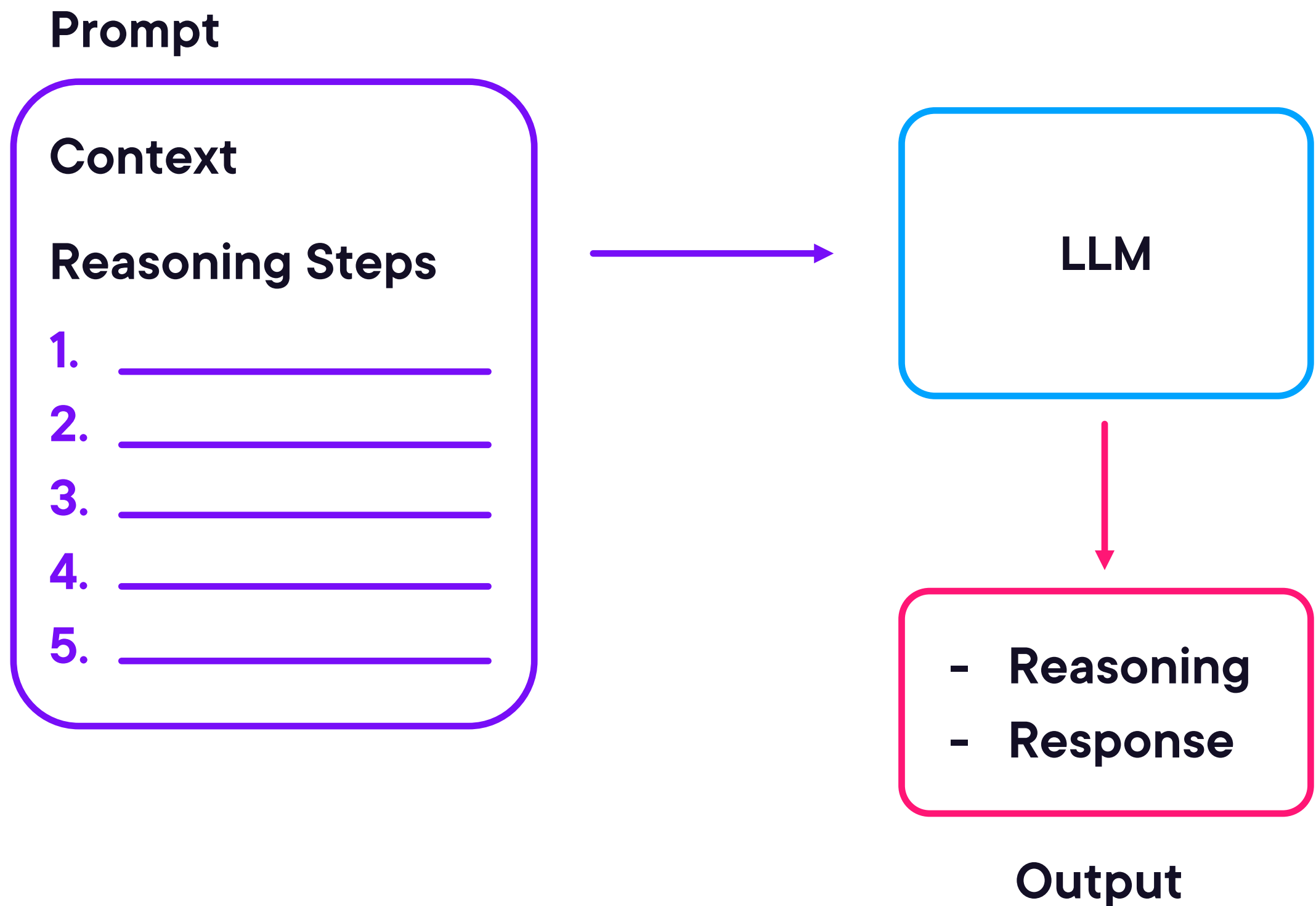




Chain of Thought Prompting



Chain-of-Thought Prompting



Prompt

You are an AI assistant that helps customers book flights on the Kesha Air airline.

Follow these steps to answer customer queries:

1. _____
2. _____
3. _____

Series of Steps

You are an AI assistant that helps customers book flights on the Kesha Air airline.

Follow these steps to answer the customer queries.

Step 1: First decide whether the user is asking a question about a specific airline or flight route.

Step 2: If the user is asking about specific flight routes, identify whether the flight routes are in the following list.

Step 3: If the message contains flight routes in the list above, list any assumptions that the user is making in their message.



**Chain-of-thought
discourages the model from
generating quick answers.**



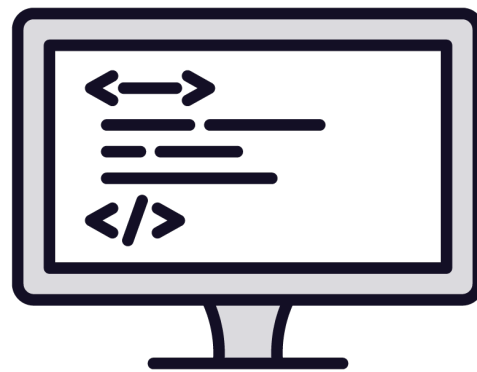
Demo



Chain of Thought Prompting

- Simple examples
- Complex examples





Hands-on Challenge

Improve the Weekly Meal Planning prompt

Take inventory of your pantry, fridge, and freezer. Include the ingredients you have available in the prompt to tailor the model's response to what you have.



Summary



Chain of Thought Prompting

- Provides specific steps
- Improves model's performance



Up Next:

Knowledge Augmentation





Knowledge Augmentation



Overview

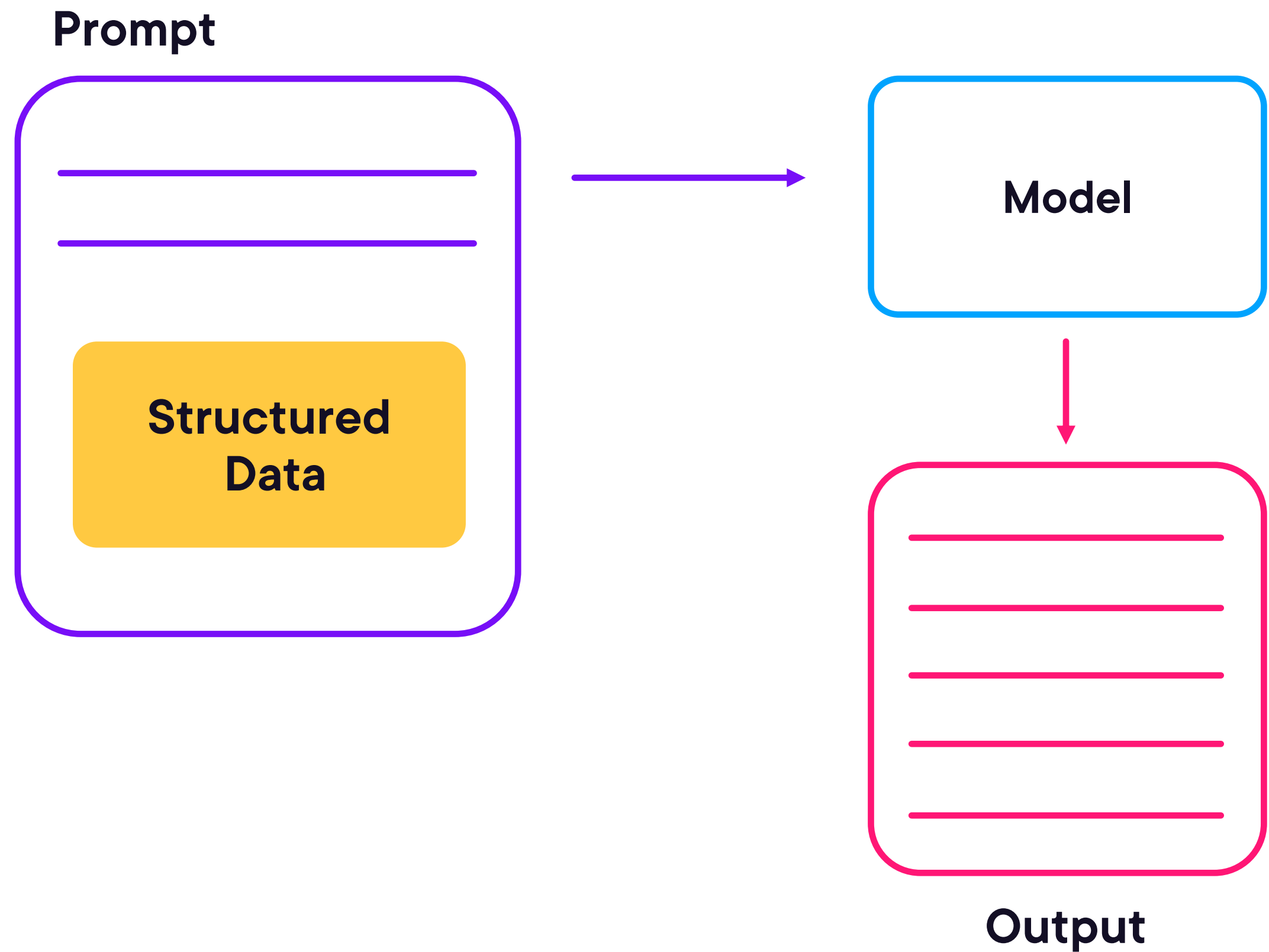


Augmenting prompts

- Relevant keywords
- Structured data
- Additional context
- External data



Prompt Augmentation



Combine with Chain-of-Thought Prompting

Prompt

You are an AI assistant that helps customers book flights on the Kesha Air airline. Follow these steps to answer customer queries:

1. _____
2. _____
3. _____

Structured Data

Flight routes:

- ATL to MIA
- LGA to LAX
- SEA to AUS



Augmentation Techniques

**Add keywords and
phrases**

**Incorporate
structured data**

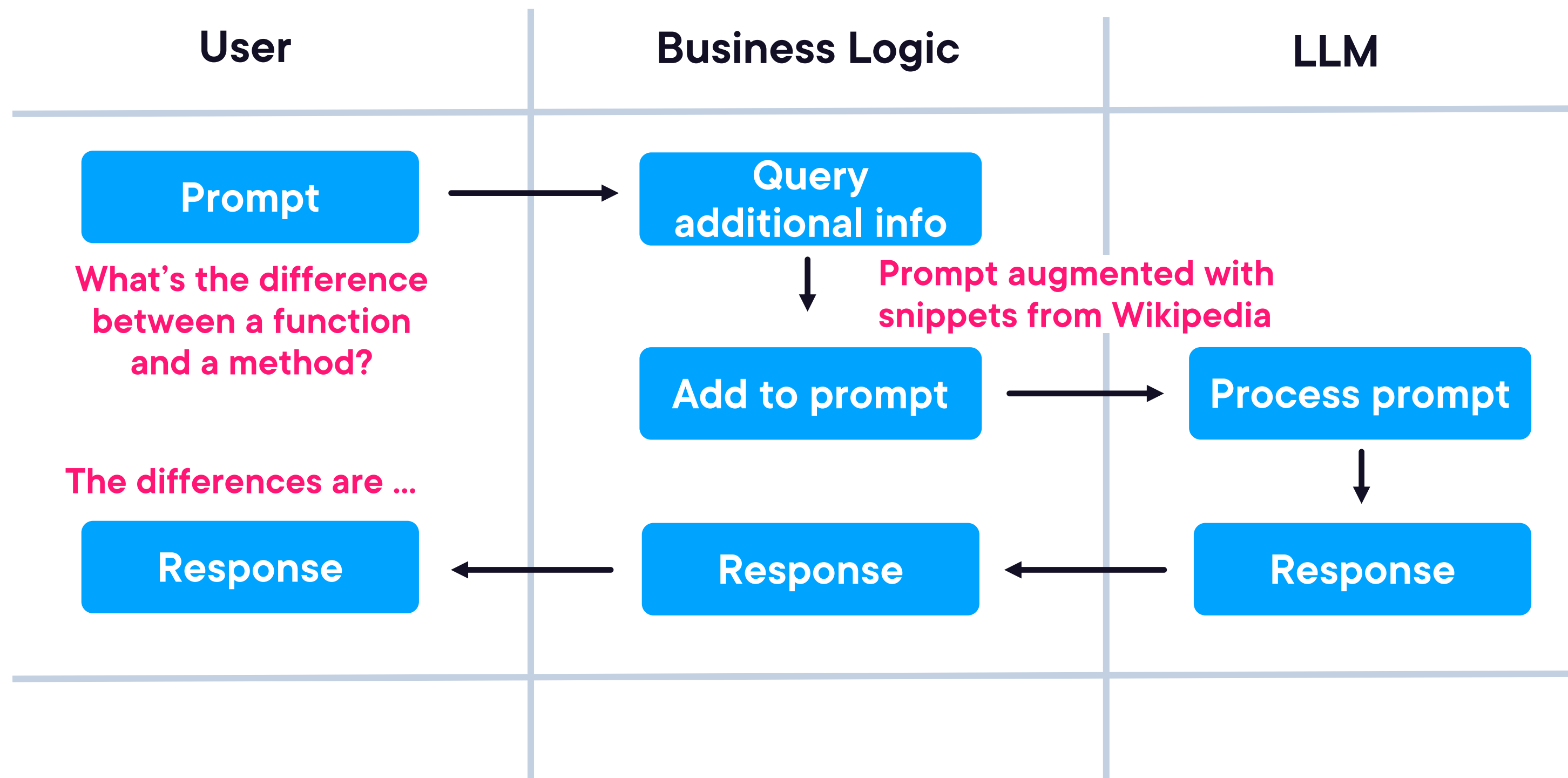
**Provide additional
context**

**Access a vector
database**

**Incorporate search
results**



Coding Tutor Application



Demo

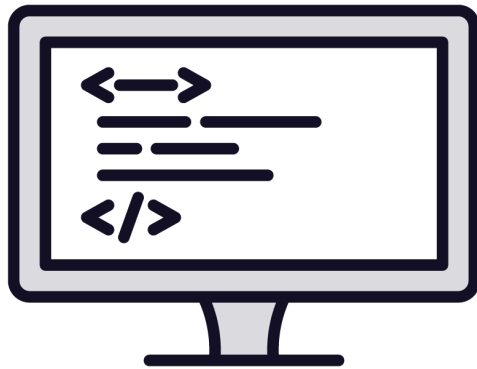


Knowledge Augmentation

- Incorporate structured data
 - Help customers book airline tickets



Turn salary negotiation prompt into a chatbot with multi-turn conversations



Tips

- Prompt the user and collect their responses using Python's `input()` function
- Each interaction with the AI assistant is a new session so the entire history of **prompts** and **responses** must be included in each exchange



Summary



Augmenting prompts

- Relevant keywords
- Structured data
- Additional context
- External data



Up Next:

Prompt Tuning





Prompt Tuning



Comparison

Prompt Tuning

Prompt tuning focuses on refining and optimizing an existing prompt to improve the model's output.

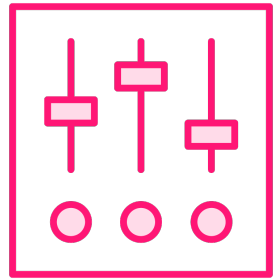
VS

Prompt Engineering

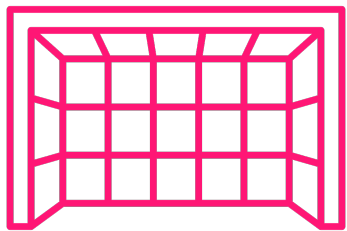
Prompt engineering involves creating and designing a prompt from scratch.



Forms of Prompt Tuning



Manual or automatic tuning



Scoring functions and optimization algorithms



Prompt variations, keyword modifications, adding context, etc.



Automatic Prompt Tuning

Define the objective

Start with initial
prompts

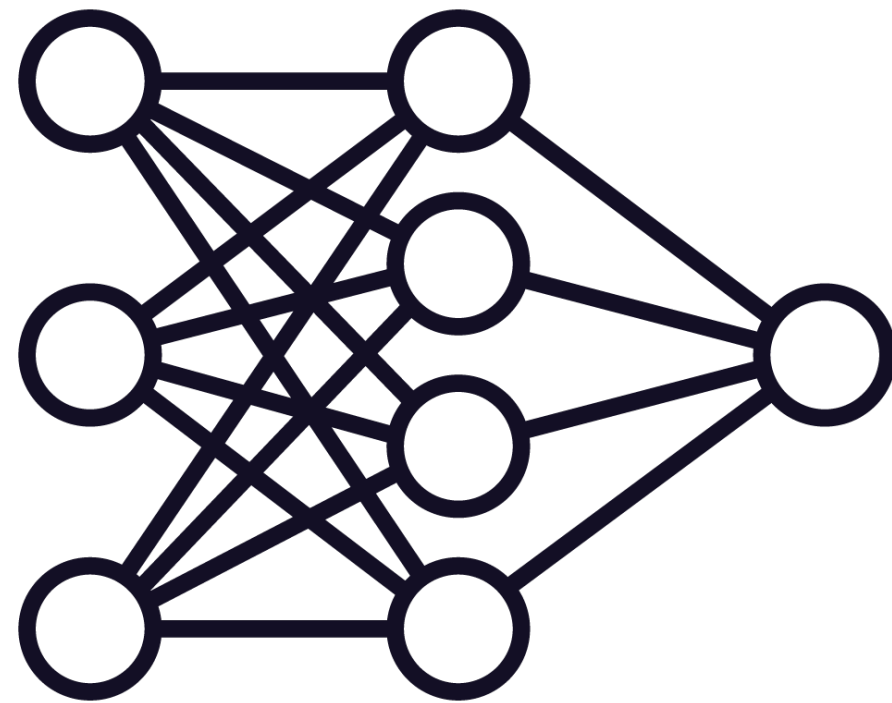
Generate responses

Evaluate responses

Update prompts

Iterate





Modify Prompt Embeddings

Automatic tuning requires access to a self-hosted LLM.



Demo



Manual Prompt Tuning Techniques

- Tuning your prompt improves performance
- Tuning is performed by changing keywords and adding additional context





Hands-on Challenge

Write a prompt that is responsible for tuning your prompt.



Summary



Prompt Tuning Overview

Prompt Tuning vs Prompt Engineering

Manual versus Automatic Prompt Tuning



Up Next:

Optimize Prompts for Specific Use Cases



Conclusion



Chain of Thought Prompting

Knowledge Augmentation

Prompt Tuning



Up Next:

Optimize Prompts for Specific Use Cases

